SOFTWARE ANALYSIS DOCUMENT (DYNAMIC VIEWS)

PROJECT: Traduce Señas

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Introduction

This document presents the analysis of the Traduce Señas software from its dynamic view, that is, it describes how the system behaves over time: which messages its components exchange, which activities are executed, through which states its main objects pass, and how the microservices collaborate to fulfill each requirement.

the Unified Modeling Language (UML) is used, following the notation of Booch et al. (2005). Use case, sequence, activity, state machine, and communication diagrams are included. All diagrams are based on the requirements of the specification report and on the microservices architecture implemented in the backend repository.

Purpose of the document

To describe the dynamic behavior of the system's critical processes so that the development team, the instructor, the evaluators, and anyone else can understand how each requirement is resolved before reviewing the code.

Scope of the analysis

The analysis covers the processes with the greatest value and risk: login, registration with email verification, password recovery, live sign translation, and capture of training samples. Simple query processes (view alphabet, view history, view statistics) follow the same request-response pattern and are summarized in tables.

Context of the analysis

Technically, the system consists of a client built with React Native and Expo (running on Android, iOS, and the web browser) and a microservices backend built with Node.js, Express, and TypeScript: an API Gateway and five domain services (iam, recognition, lexicon, analytics, and profile), each with its own PostgreSQL database, MongoDB for training samples, and RabbitMQ for asynchronous event-based communication.

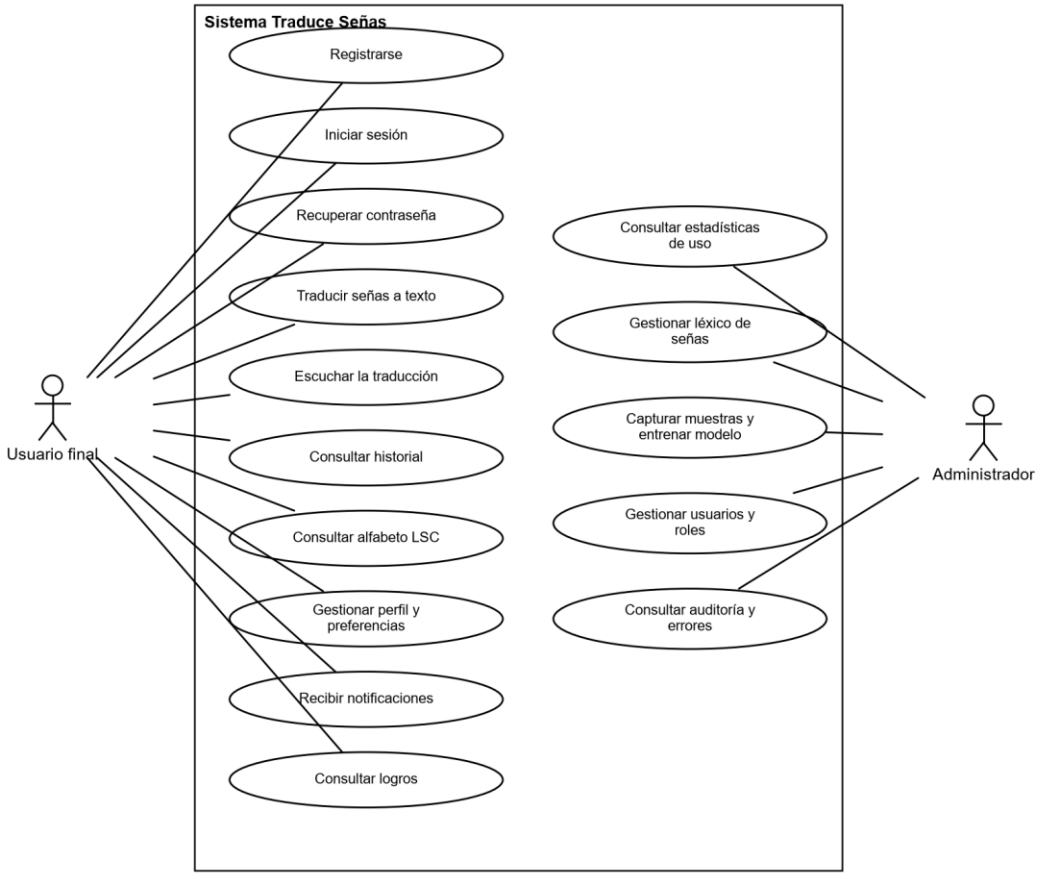
Table 1
Participants appearing in the dynamic diagrams

Participant	Type	Responsibility in the behavior
End user	Actor	Deaf or hearing person who uses the application
Administrator	Actor	Manages the lexicon, users, and samples, and queries statistics
App (React Native / Expo)	Client	Interface, camera, local recognition with MediaPipe, and REST calls
MediaPipe	On-device library	Detects the hand and returns 21 points per video frame
api-gateway	Edge service	Single entry point: validates JWT, limits requests, and routes
iam-service	Microservice	Accounts, authentication, roles, and password recovery
recognition-service	Microservice	Translations, sign templates, and training samples
lexicon-service	Microservice	Sign catalog, categories, and alphabet
analytics-service	Microservice	Recording of usage events and statistics
profile-service	Microservice	Profile, preferences, notifications, and achievements
RabbitMQ	Message broker	Transports domain events between services

Use case model

The use case diagram defines the boundaries of the actors' interactions with the system and is the starting point for the other dynamic views. The administrator inherits all the capabilities of the end user and also has access to the management functions.

Figure 1
System use case diagram



Note. Own elaboration.

Table 2
Relationship between use cases and dynamic diagrams

Use case	Requirement	Diagrams that describe it
CU-01 Log in	RF1	Sequence (Figure 2), account states (Figure 9)
CU-02 Register	RF1	Sequence (Figure 3), activities (Figure 8)
CU-03 Recover password	RF1	Sequence (Figure 4)
CU-04 Translate signs to text	RF3	Sequence (Figure 5), activities (Figure 7), states (Figure 10), communication (Figure 11)
CU-05 Capture samples and train	RF3	Sequence (Figure 6)

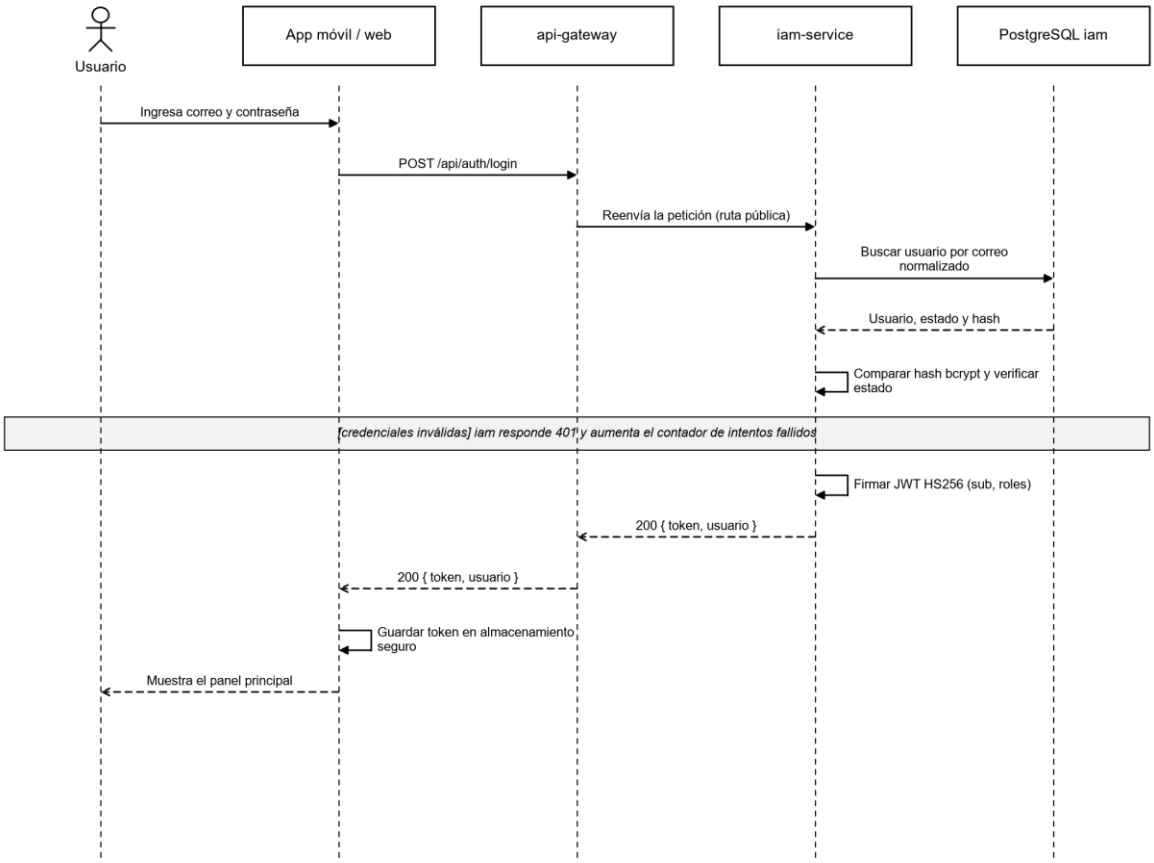
Sequence diagrams

Sequence diagrams show the temporal order of messages between participants. Solid arrows are calls; dashed arrows are responses. Gray boxes indicate conditions or repetitions.

Log in

The client never calls a microservice directly: all requests go through the api-gateway on port 8080. The /api/auth route is public, so the gateway forwards it without requiring a token. The iam-service compares the password hash with bcrypt, verifies that the account is active, and signs a JWT that includes the user identifier and their roles, so that no other service has to query the iam database (Jones et al., 2015).

Figure 2
Sequence diagram — Log in



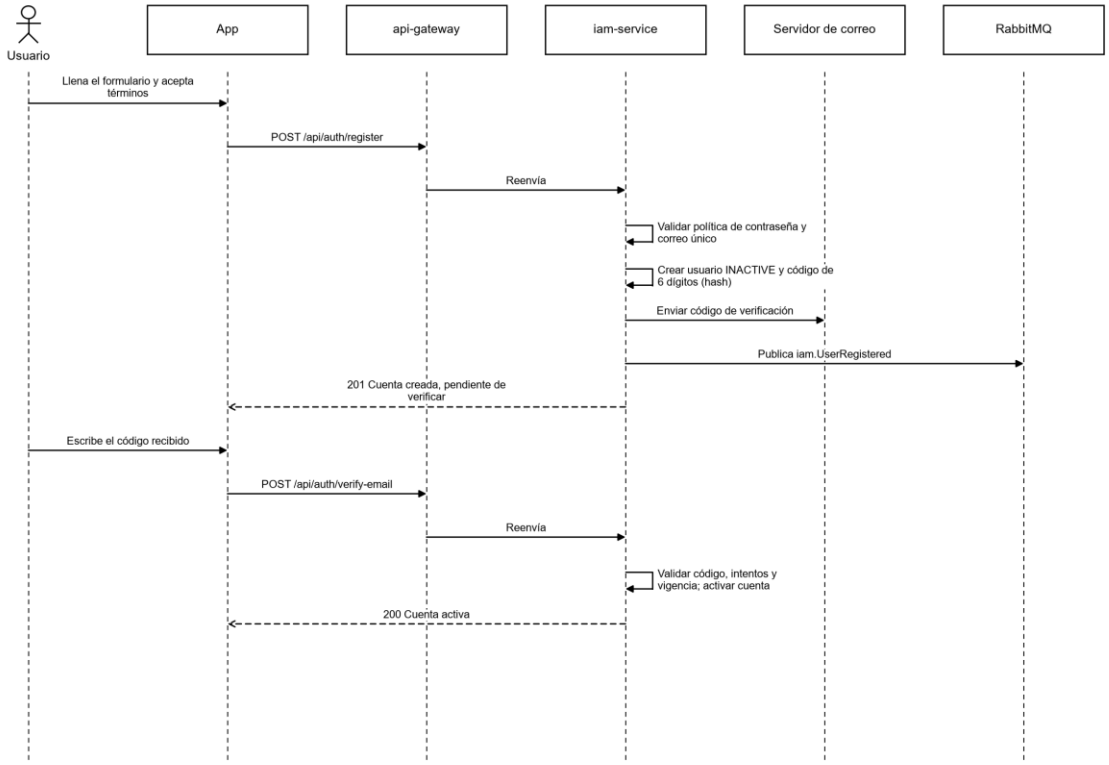
Note. Own elaboration.

Registering and verifying the email

The account is created in the INACTIVE state and is only activated when the user enters the six-digit code sent to their email. The code is stored as a hash, expires, and counts failed attempts. When the account is created, the iam-service publishes the UserRegistered event so

that the profile-service creates the profile without a direct call between services.

Figure 3
Sequence diagram — Registration and email verification

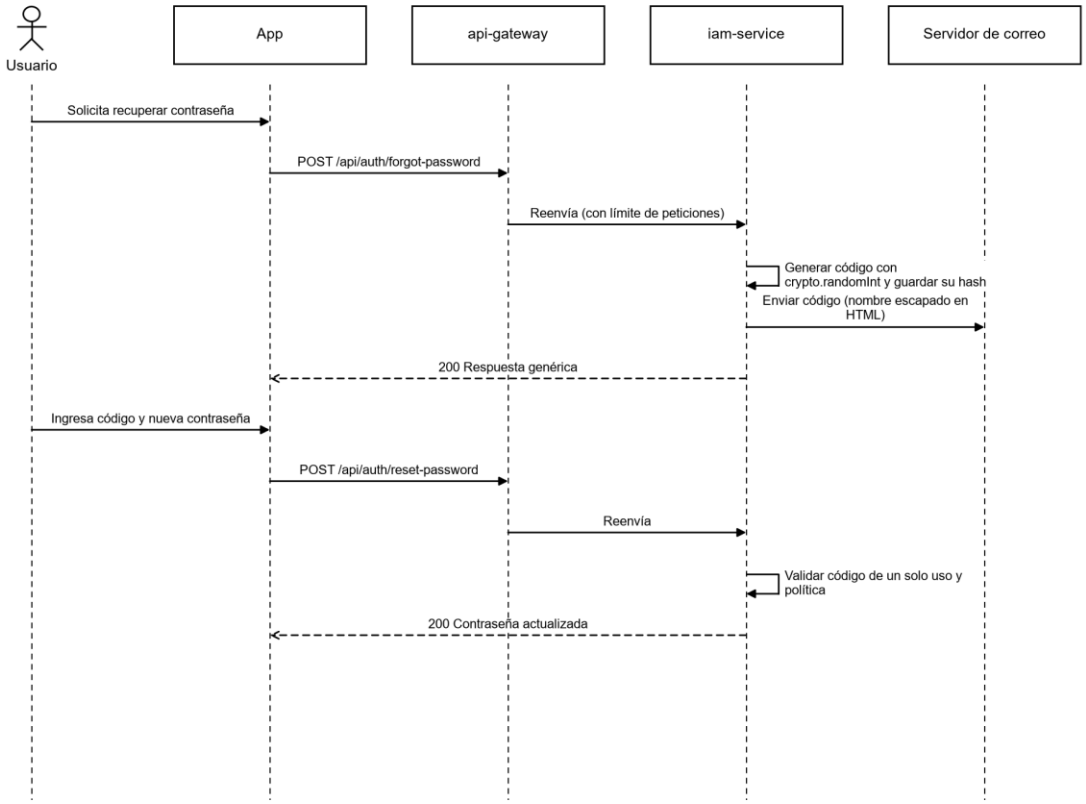


Note. Own elaboration.

Recovering the password

The response to the request is generic so as not to reveal whether an email exists. The code is generated with crypto.randomInt, has an attempt limit, and the /api/auth route has a strict request limit to prevent brute-force attacks.

Figure 4
Sequence diagram — Password recovery

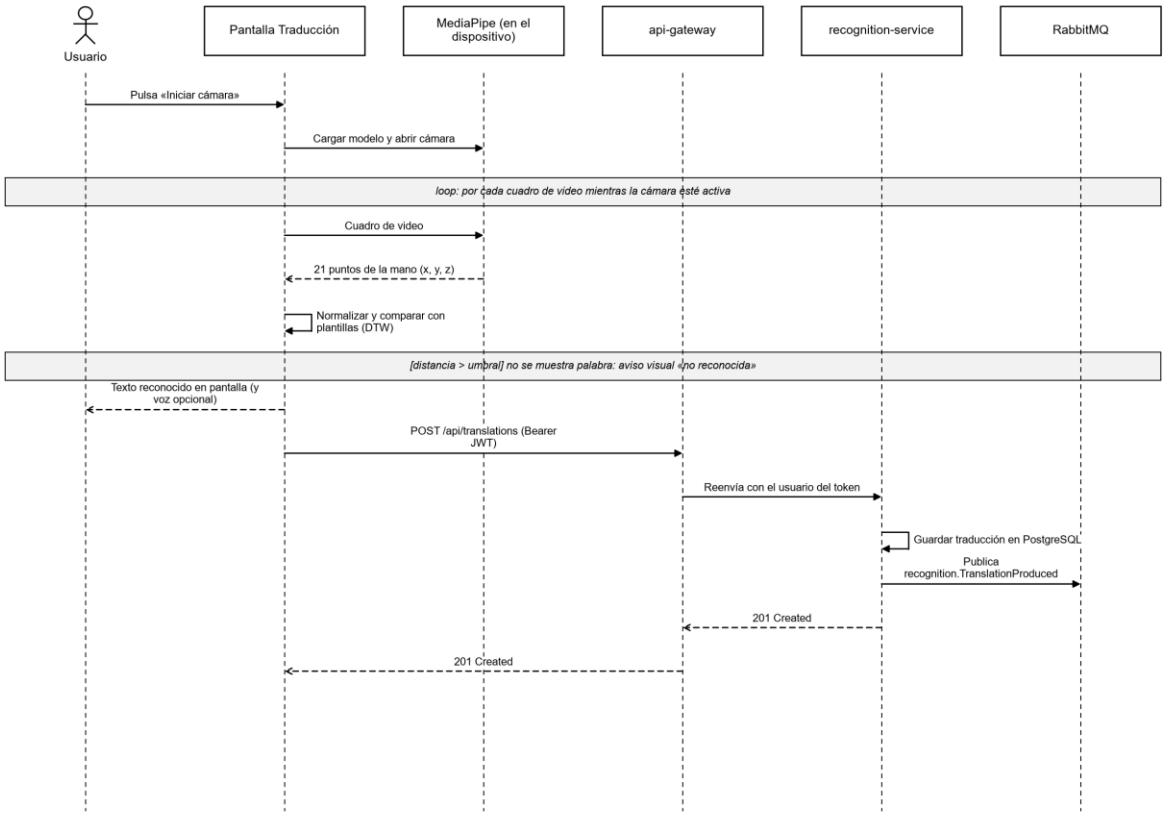


Note. Own elaboration.

Translating signs live

Recognition runs on the device: MediaPipe obtains the 21 hand points in each frame, the application normalizes them (centers them on the wrist and scales them by the size of the palm), and compares them with the templates using Dynamic Time Warping (Sakoe and Chiba, 1978). If the distance exceeds the threshold, the application does not guess: it displays a visual warning. Only the final result is sent to the backend to be saved in the history, which reduces latency and protects privacy because the video never leaves the device.

Figure 5
Sequence diagram — Live translation

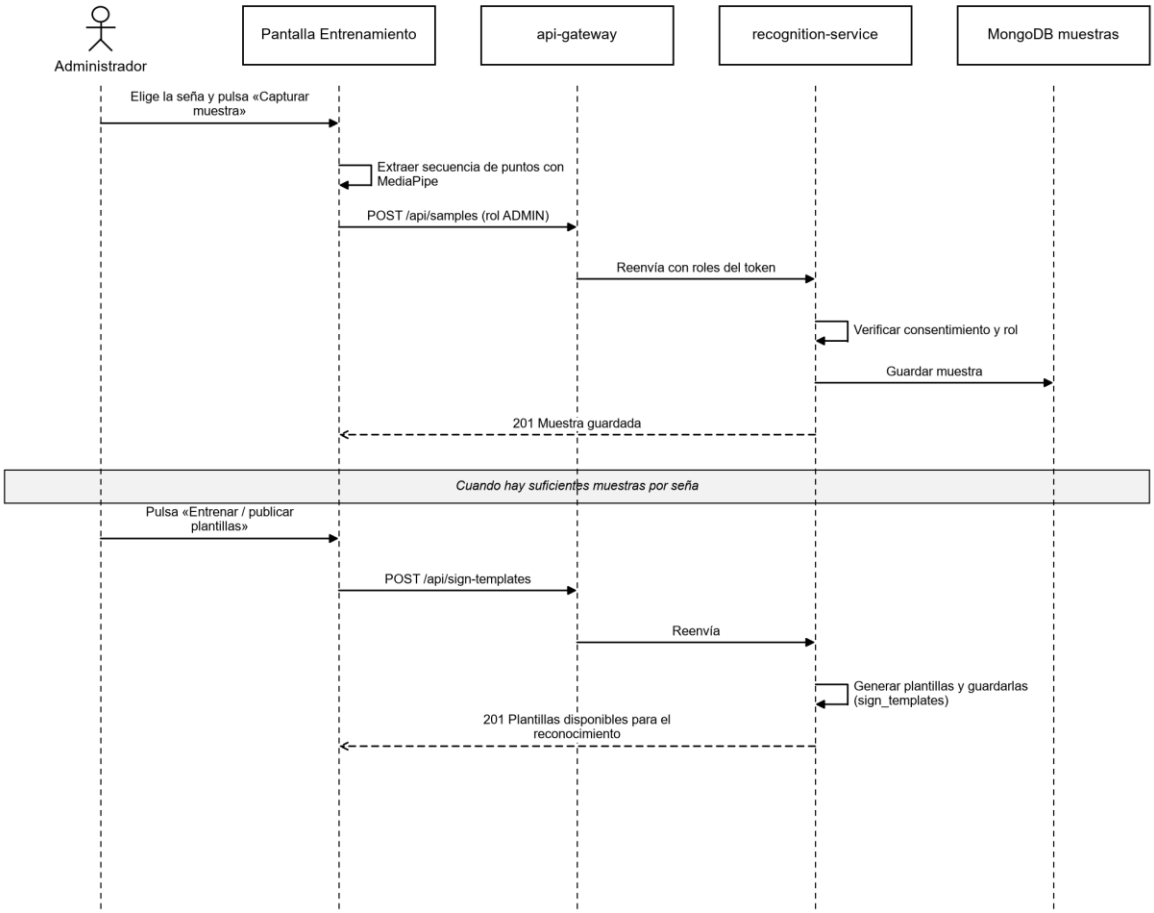


Note. Own elaboration.

Capturing samples and training the recognition

This flow is restricted to the ADMIN role. Each sample requires demonstrable consent from the signer (Law 1581 of 2012) and is stored in MongoDB, separate from the relational data. With enough samples, the templates are generated, which the application then downloads.

Figure 6
Sequence diagram — Sample capture and training



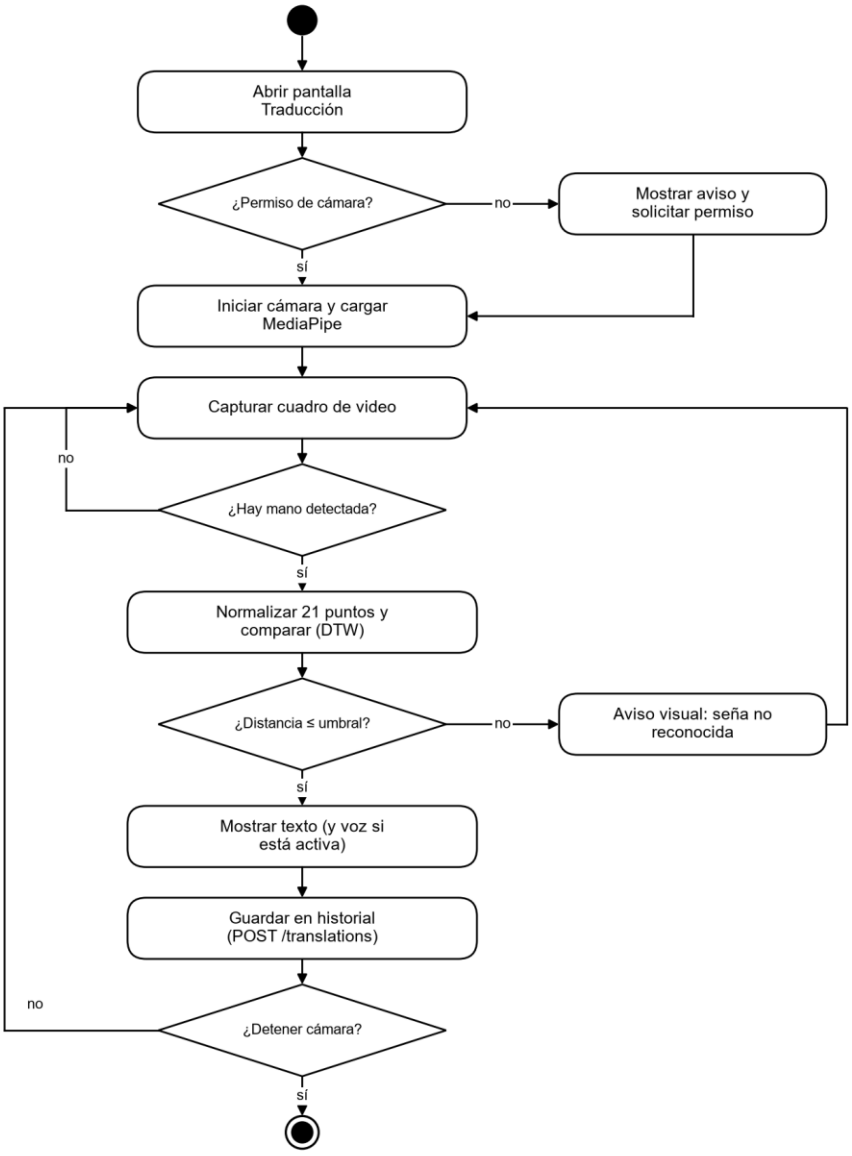
Note. Own elaboration.

Activity diagrams

Activity diagrams describe the workflow of a process, including its decisions and cycles.

Translation process

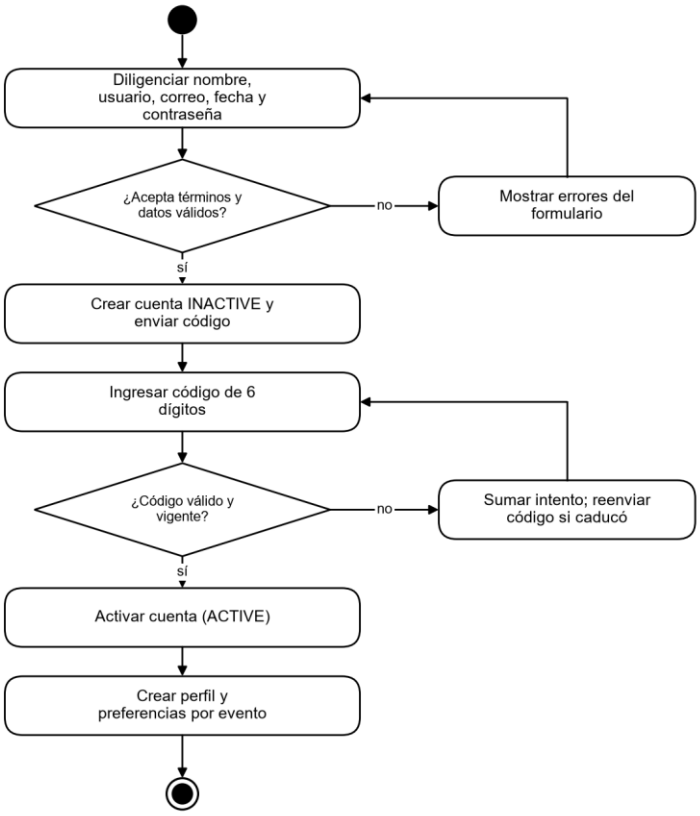
Figure 7
Activity diagram — Sign translation



Note. Own elaboration.

Registration process

Figure 8
Activity diagram — User registration



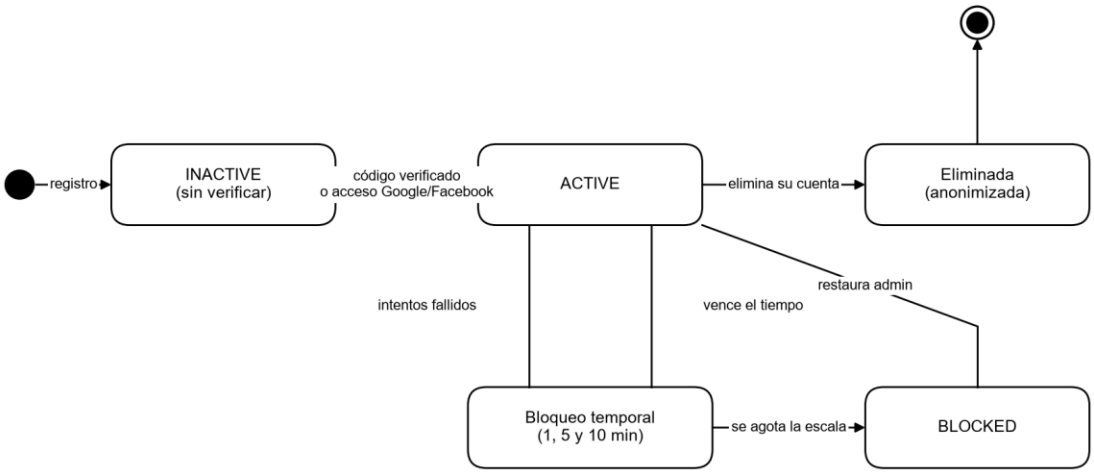
Note. Own elaboration.

State machine diagrams

User account states

The user account is the object with the most important lifecycle for security. A temporary lock is a date that lifts itself; when the lock scale is exhausted, the account moves to BLOCKED, and only an administrator can restore it. Deletion anonymizes the data instead of physically erasing it, to preserve audit integrity and comply with the right to erasure.

Figure 9
State diagram — User account



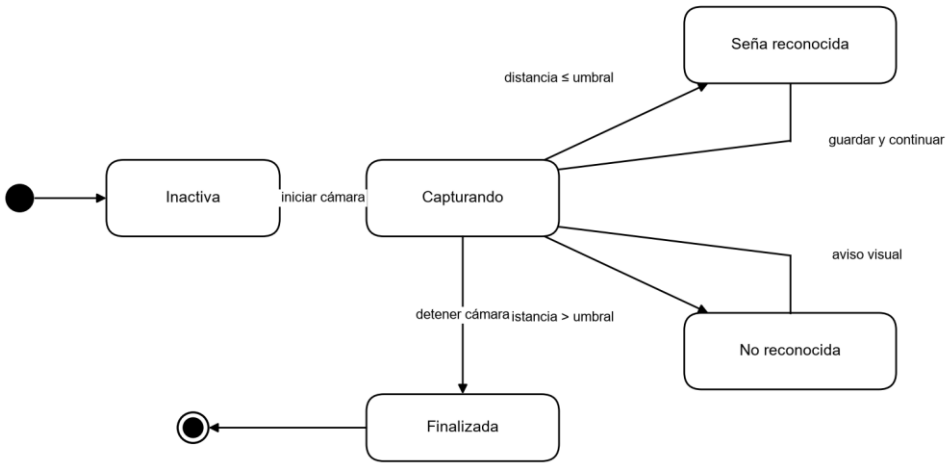
Note. Own elaboration.

Table 3
User account transitions

Source state	Event	Target state	Rule
—	Registration with email	INACTIVE	Must accept terms (FR-002)
—	Registration with Google / Facebook	ACTIVE	The provider has already verified the email (FR-008)
INACTIVE	Valid verification code	ACTIVE	Single-use and valid code (FR-004)
ACTIVE	Repeated failed attempts	Temporary lock	Waits of 1, 5, and 10 minutes (FR-006)
Temporary lock	Lock scale exhausted	BLOCKED	Only an administrator restores (FR-007)
ACTIVE	The user deletes their account	Deleted	The row is anonymized (FR-014)

States of a translation session

Figure 10
State diagram — Translation session

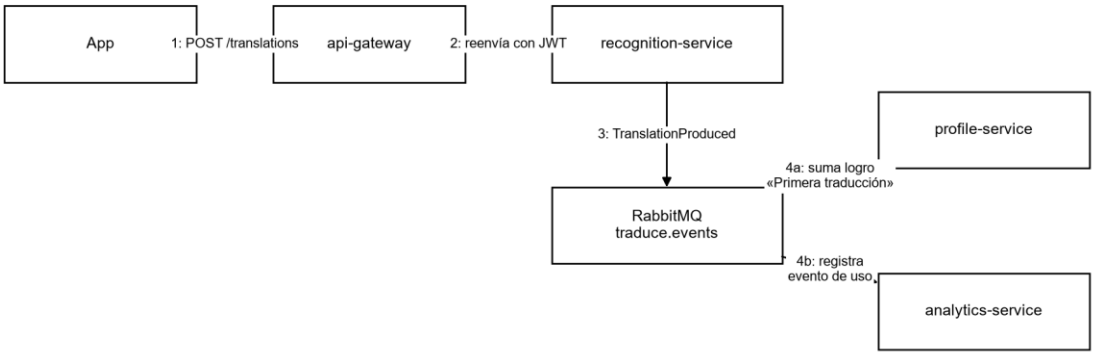


Note. Own elaboration.

Communication diagram between services

The communication diagram shows how the objects collaborate to complete a task, numbering the messages. When a translation is saved, the recognition-service publishes TranslationProduced to the traduce.events exchange. The profile-service consumes it to update achievements (for example, “First translation”) and the analytics-service records it as a usage event. No service waits for another's response to complete the user's request, which reduces coupling (Newman, 2021).

Figure 11
Communication diagram — Saved translation and derived events



Note. Own elaboration.

Table 4
Main domain events

Event	Publisher	Consumers	Effect
iam.UserRegistered	iam-service	profile, analytics	Creates profile and preferences; counts the registration
iam.AccountDeletionRequested	iam-service	recognition, profile	Anonymizes history and profile data
recognition.TranslationProduced	recognition-service	profile, analytics	Advances achievements; records usage
lexicon.SignPublished	lexicon-service	recognition, analytics	Updates the catalog of recognizable signs
lexicon.SignWithdrawn	lexicon-service	recognition	Removes the sign from recognition

Summary of query interactions

Table 5
Query request-response interactions

Interaction	Route (via gateway)	Service	Access
View alphabet	GET /api/lexicon/alphabet	lexicon-service	Public
Search signs	GET /api/lexicon/search	lexicon-service	Public
View history	GET /api/translations/history	recognition-service	Authenticated user (own data only)
Delete from history	DELETE /api/translations/:id	recognition-service	Owner of the translation
View profile	GET /api/profile/me	profile-service	Authenticated user
View statistics	GET /api/analytics/summary, /daily, /top-signs	analytics-service	ADMIN role
Manage roles	POST /api/iam/users/:id/roles	iam-service	ADMIN role

Conclusions

The dynamic views confirm three decisions from the analysis: recognition is performed on the device to meet the real-time requirement and protect privacy; every request goes through a single gateway that centralizes security; and the services are coordinated through asynchronous events, so that the failure of one does not block the user's operation.

The diagrams in this document serve as a reference for the detailed design and for writing the integration and end-to-end test cases.

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